

Can rewarding task-switching increase cognitive flexibility?

Can reward shape flexibility?



Background: Task switching supports cognitive flexibility but is effortful. Recent work suggests that rewarding effortful switches increases subsequent switching, implying that switching behavior may be conditioned, similar to the conditioning of specific actions.^{1,2}



Gap: However, because these findings are based on between-subject effects, it remains unclear whether rewards promote a general tendency to switch or whether participants learned switch-reward associations.



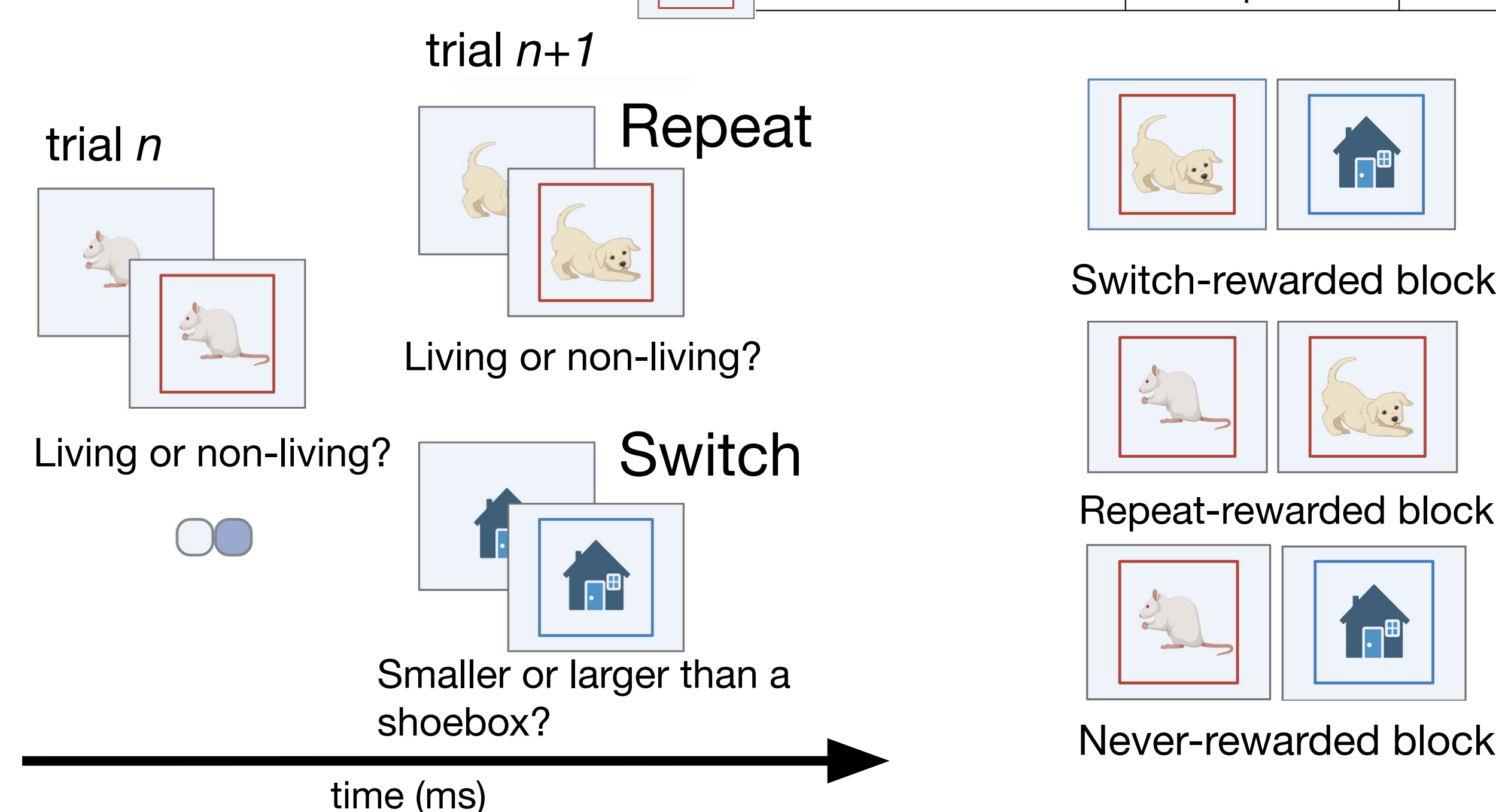
Central Question: Can switching be conditioned? To address this question, we tested switch conditioning within subjects using an item-specific conditioning procedure.

Task & Manipulation

- Seventy Purdue University undergraduates; final sample of **N=58** after exclusions. All participants provided informed consent.

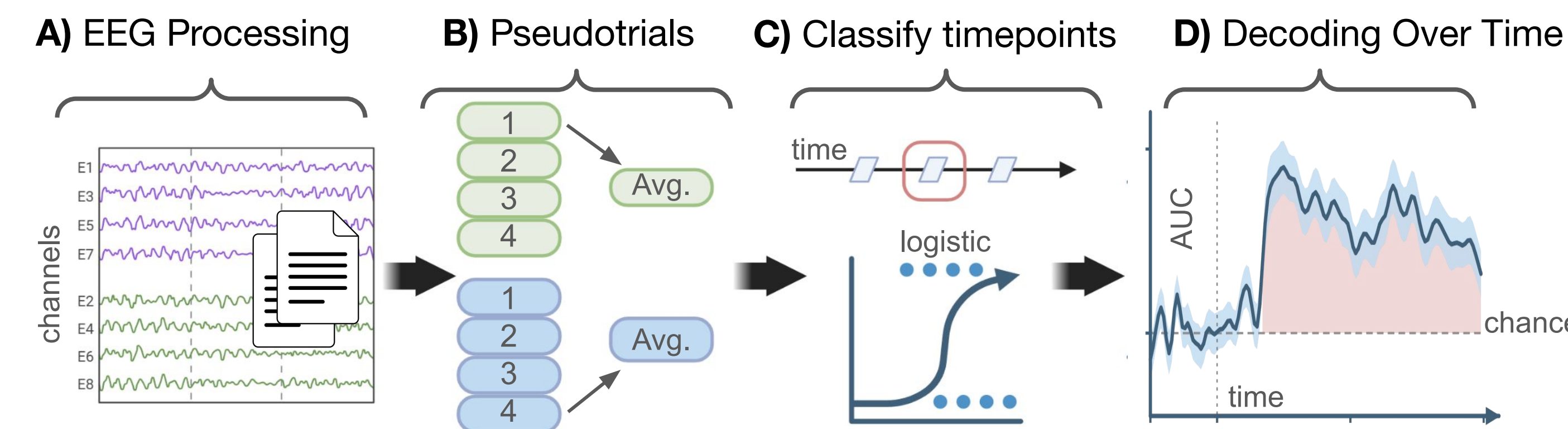
Task-Switching Paradigm

item-specific control conditioning			
condition	trial	reward	
switch-rewarded	switch	+6	
	repeat	+0	
repeat-rewarded	switch	+0	
	repeat	+6	
never-rewarded	switch	+0	
	repeat	+0	



Participants switched between categorizing items/pictures as living or nonliving or as larger or smaller than a carry-on suitcase. The required categorization task was indicated by the color of the frame (red vs. blue).

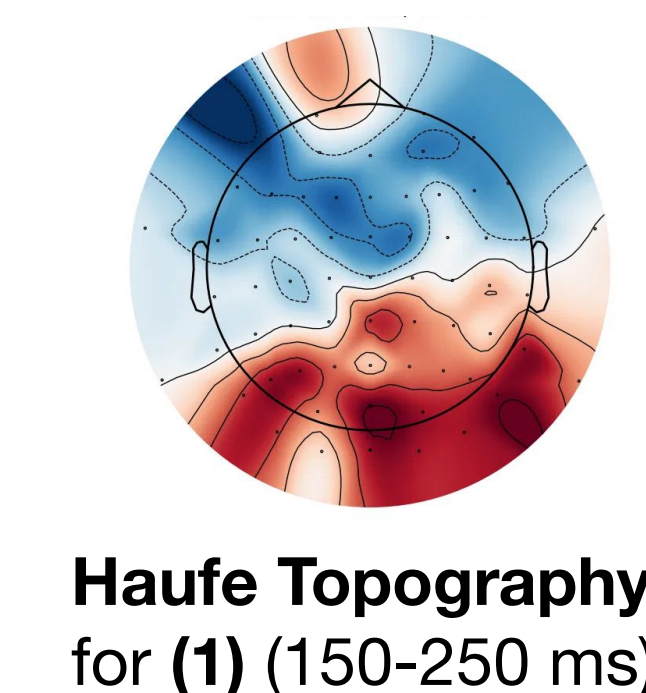
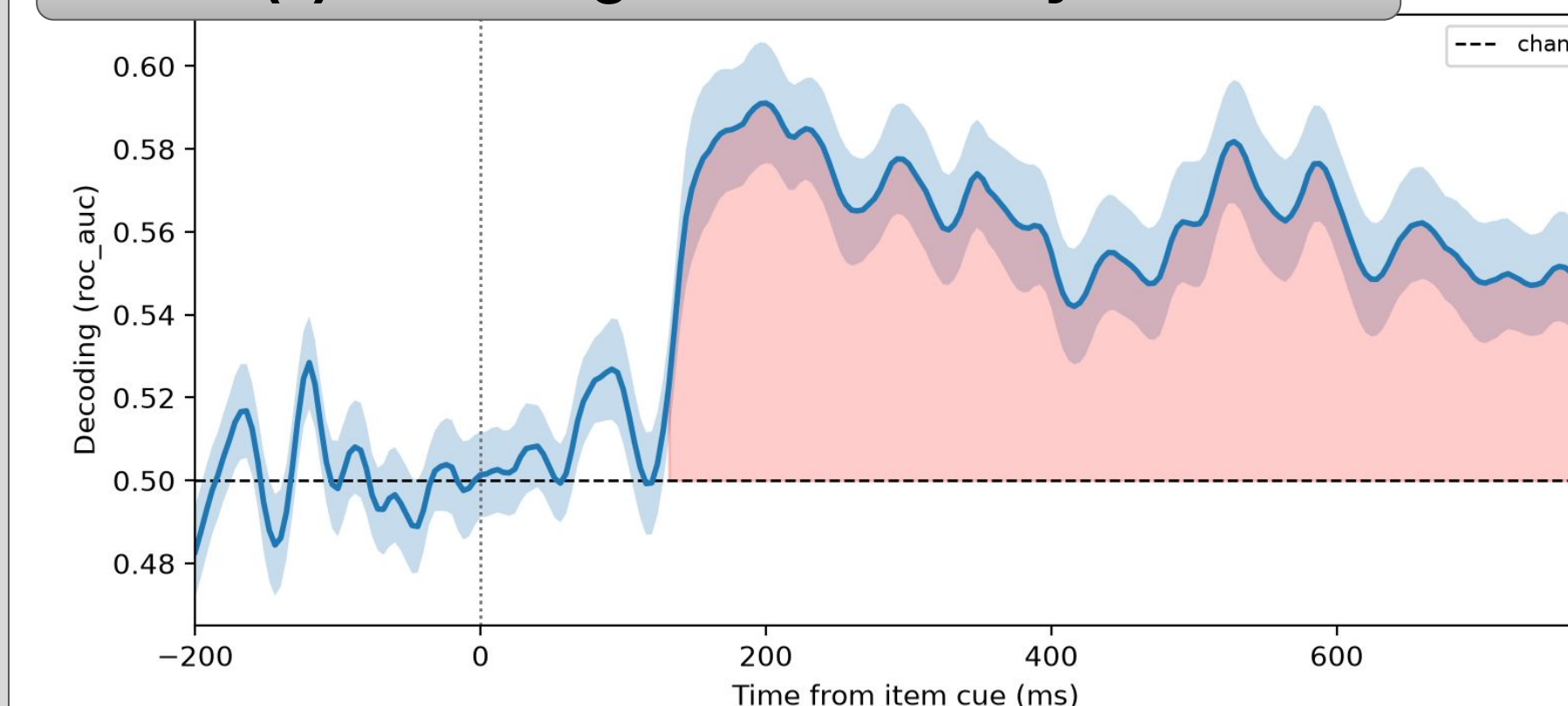
Multivariate Pattern Analysis (MVPA)



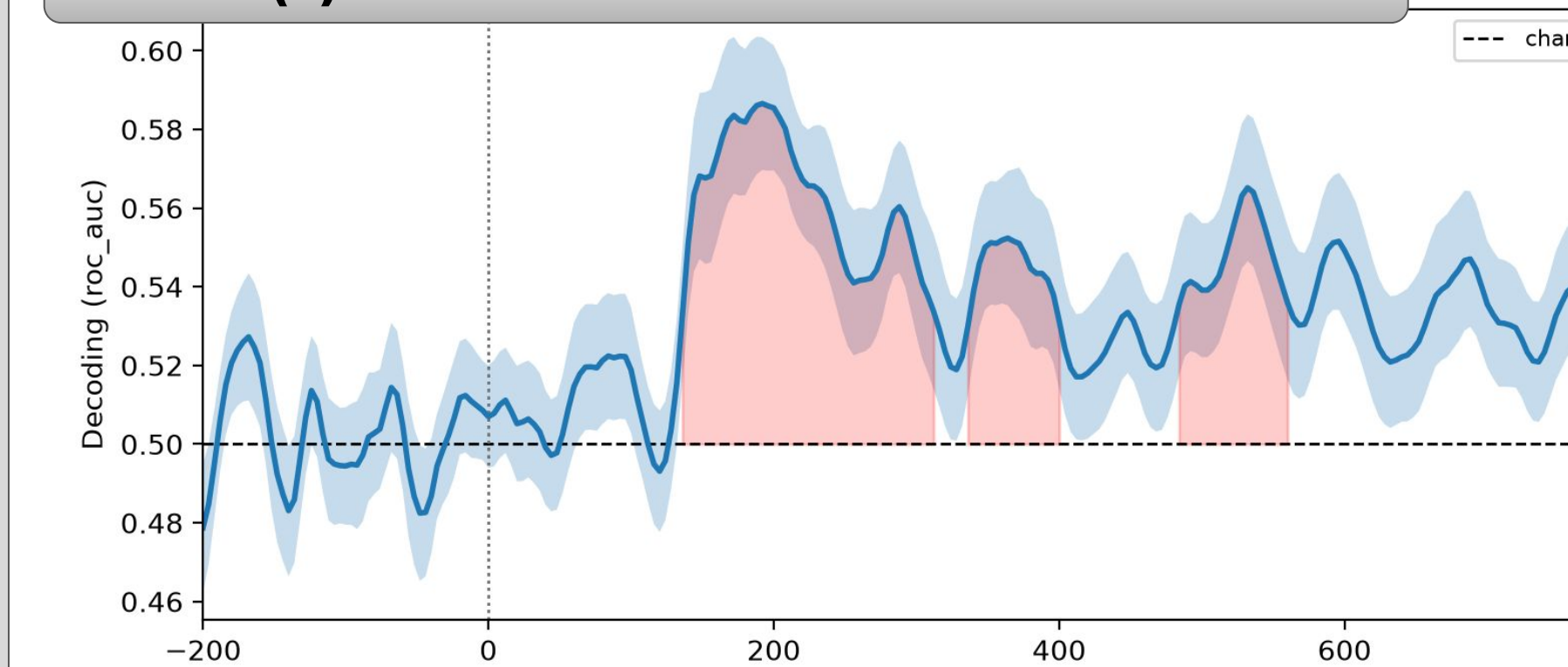
(A) 64-channel EEG, ICA-cleaned, epoched around item onset. (B) Four same-class trials averaged into a pseudotrial, redrawn 100 times. (C) A logistic classifier trained separately at each timepoint, tested on held-out trials. (D) AUC over time, tested against chance by cluster permutation (10,000 perms). Models 1 to 4 reuse decoder with different train/test splits; the TGM trains at every timepoint and tests at every other.

MVPA - Results

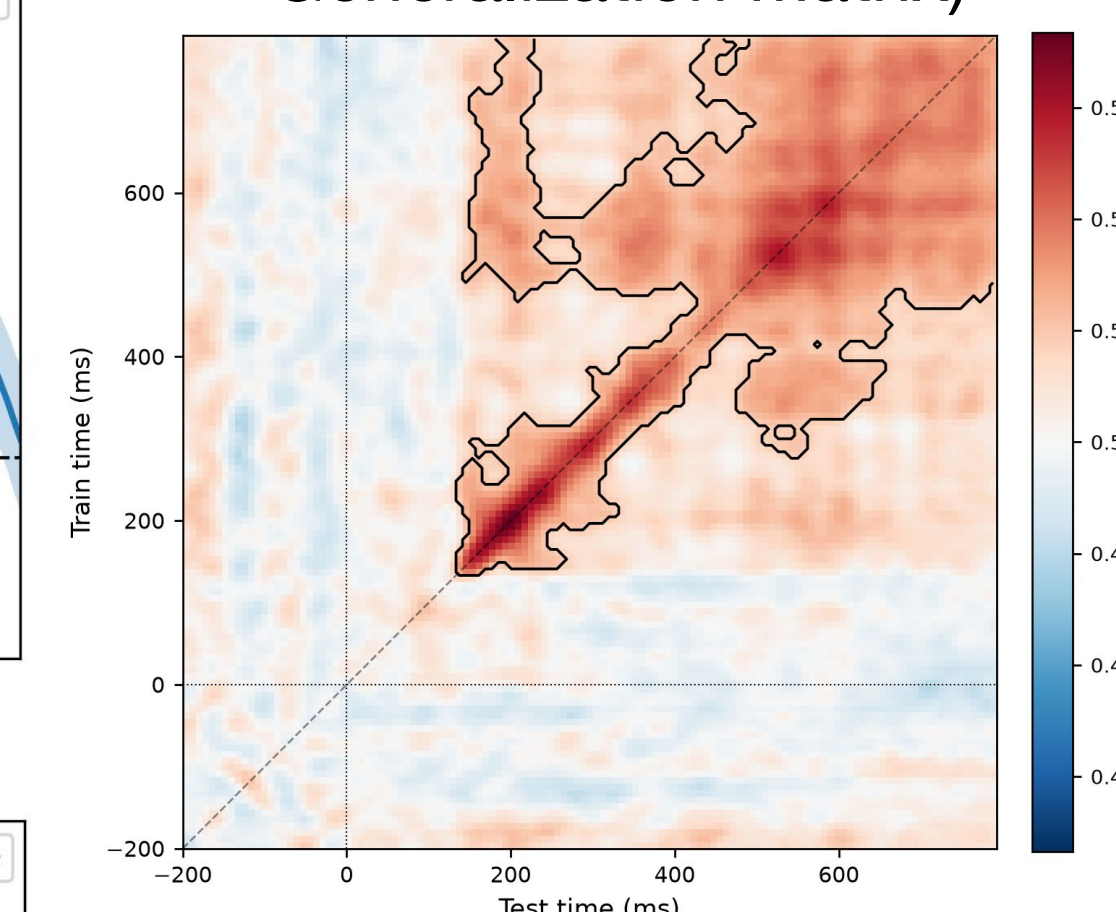
Model (1) decoding reward history



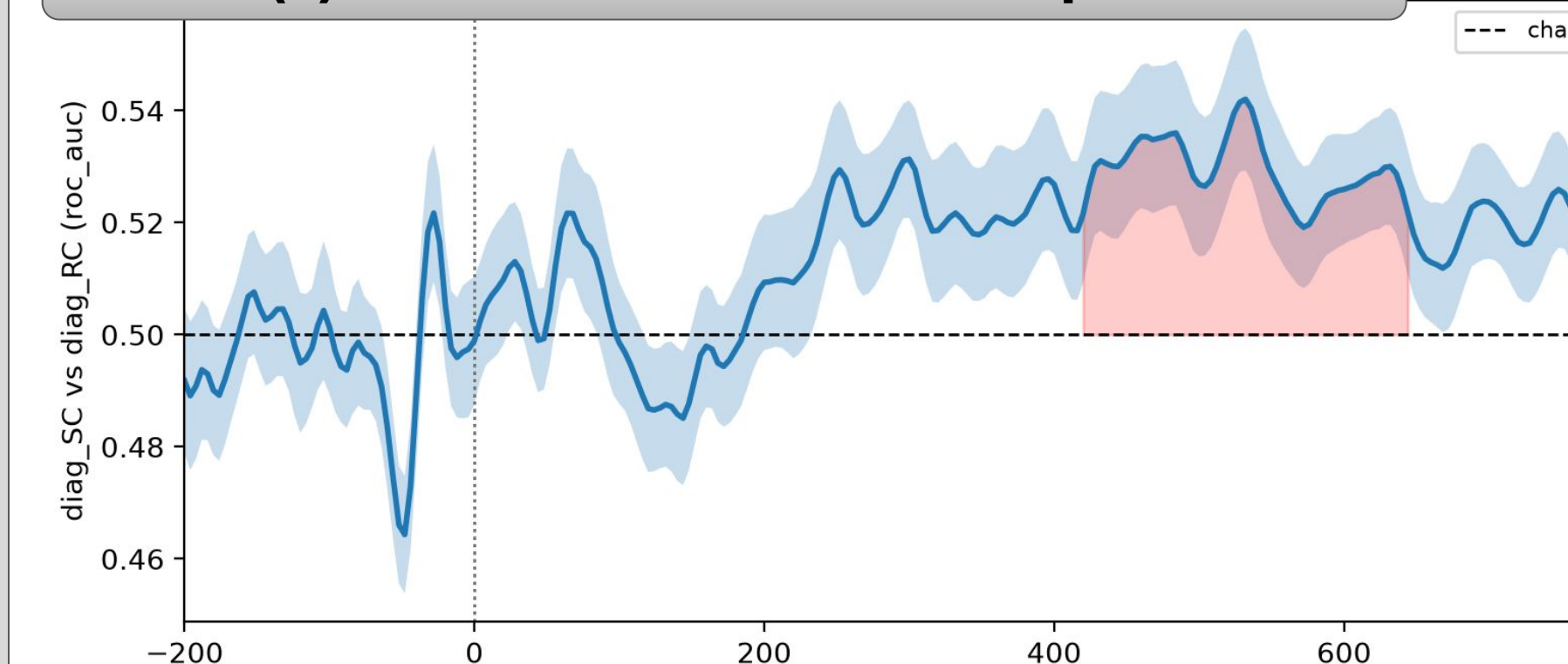
Model (2) neutral-context transfer



TGM (Temporal Generalization Matrix)

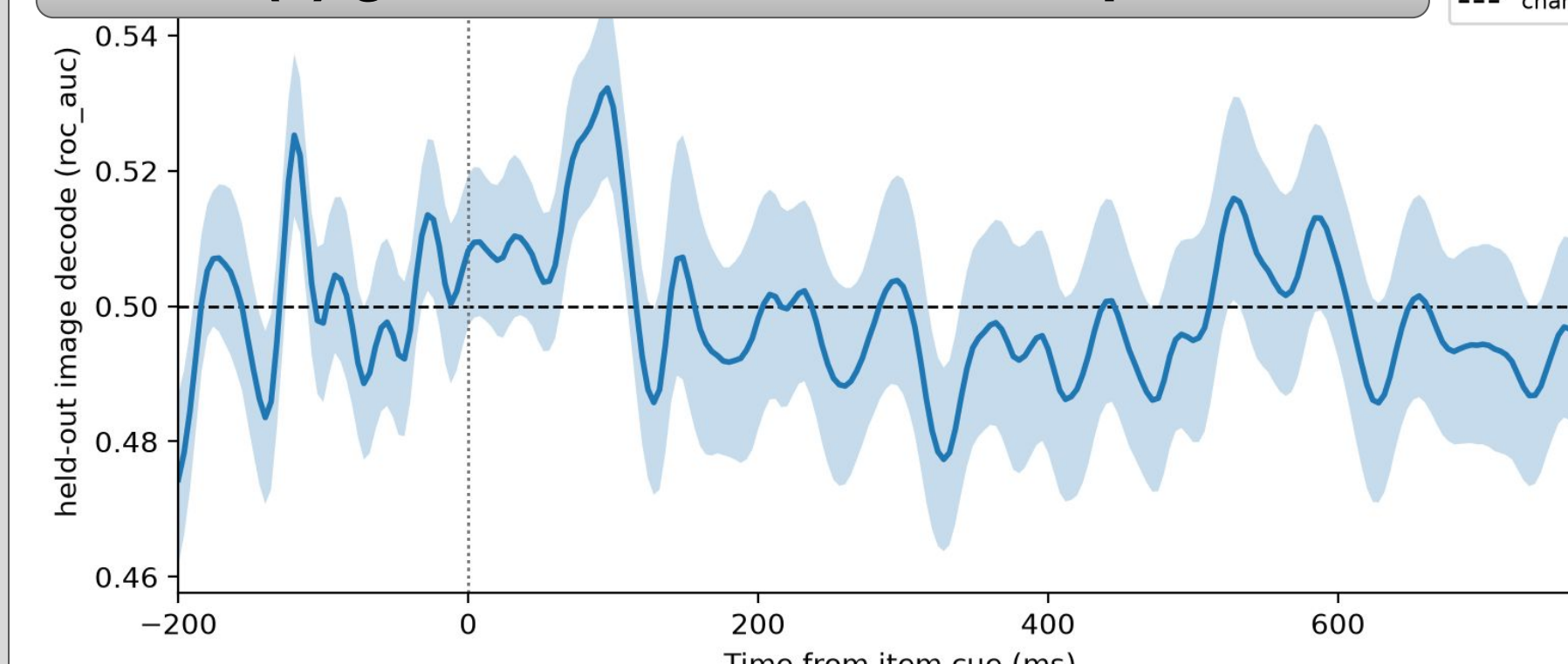


Model (3) transfer to unrewarded probes



Cluster permutation (10,000 perms); outlined area is significant (chance = .5)

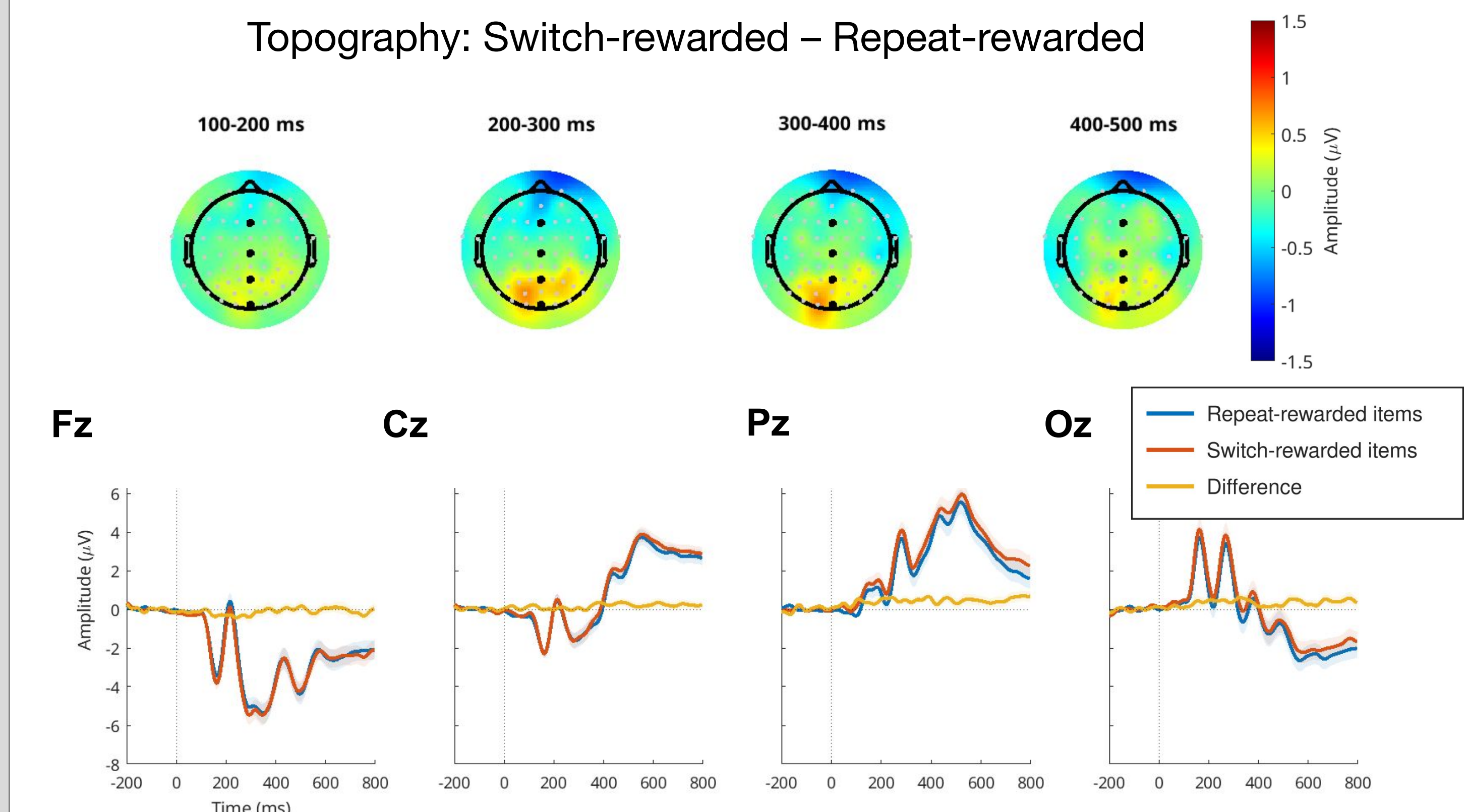
Model (4) generalization to unseen pictures



- Model (1):** 132 ms onset, peak AUC = .59
- Model (2):** peak AUC = .58, $p = .002$
- Model (3):** 420–644 ms, peak AUC = .54, $p = .002$
- Model (4):** no significance
- Haufe:** posterior positivity for switch-rewarded
- TGM:** representations shifted early before stabilizing

ERP - Methods & Results

Topography: Switch-rewarded – Repeat-rewarded



Stimulus-locked ERPs (–200–800 ms) corrected, artifact-rejected, and averaged by condition, and examined at Fz, Cz, Pz, and Oz.

Conventional ERPs from the same trials show clear visual and P3 components, confirming data quality, but the two conditions largely overlap. The condition information appears to be distributed across channels and too subtle to emerge in single channel mean amplitudes.

Key Takeaways

- Reward learning was primarily picture-specific:** brain activity distinguished pictures linked to switching versus repeating (model 1).
- The picture-specific signal persisted across contexts:** it remained detectable when switching and repeating were equally likely to be rewarded (model 2).
- Broader context mattered later:** never-rewarded pictures eventually reflected the reward state of the context in which they appeared (model 3).
- Learning did not generalize across images:** decoding failed for held-out pictures, arguing against a broad “switching is rewarding” rule and suggesting it is more item-specific (model 4).

Acknowledgements & References

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¹Braem (2017); Cognition
²Held et al. (2024); Collabra

Correspondence:
chiu56@purdue.edu